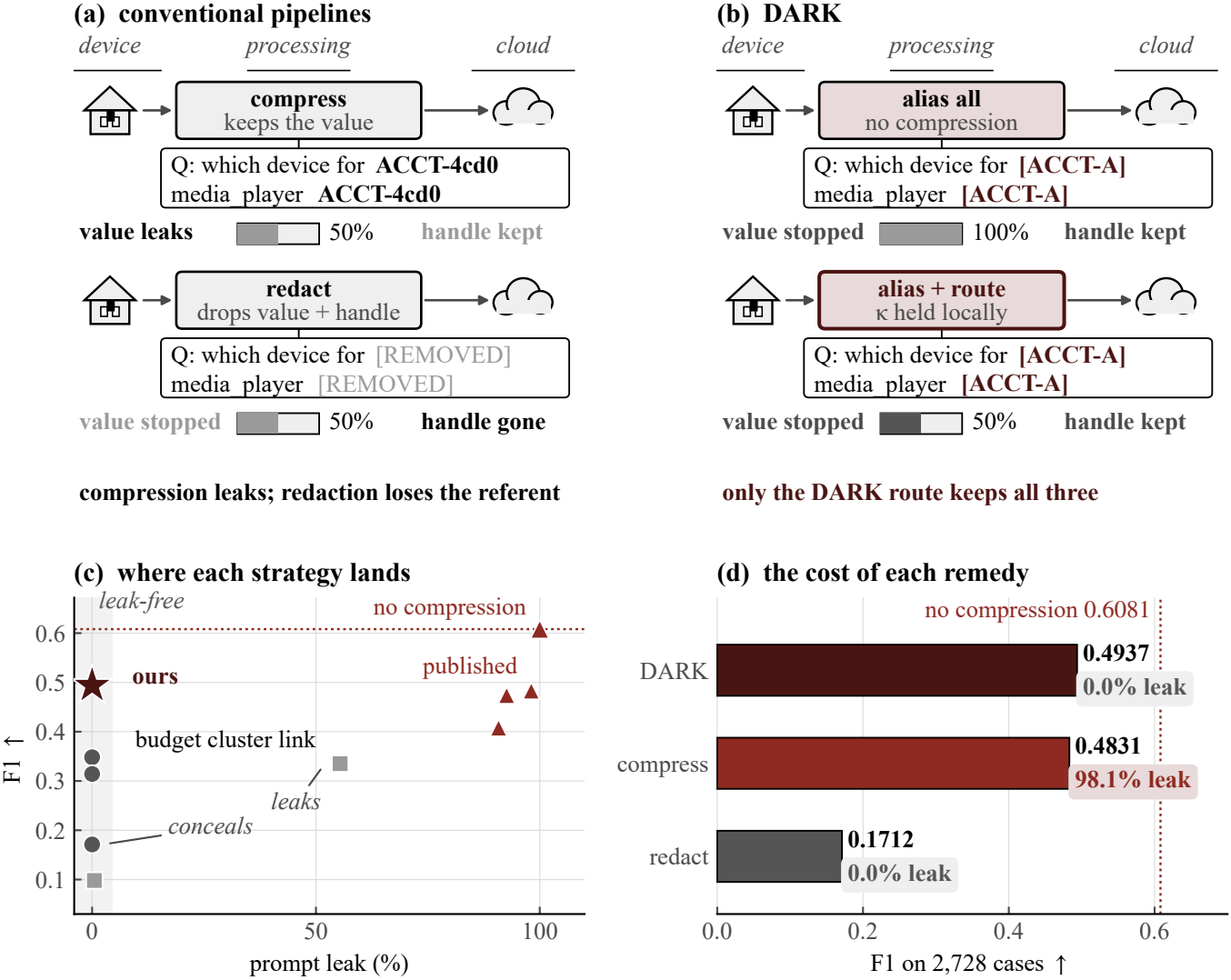

(a) conventional pipelines
device
processing
cloud
compress
keeps the value
Q: which device for ACCT-4cd0
media_player ACCT-4cd0
value leaks
50%
handle kept
redact
drops value + handle
Q: which device for [REMOVED]
media_player [REMOVED]
value stopped
50%
handle gone
compression leaks; redaction loses the referent
(b) DARK
device
processing
cloud
alias all
no compression
Q: which device for [ACCT-A]
media_player [ACCT-A]
value stopped
100%
handle kept
alias + route
κ held locally
Q: which device for [ACCT-A]
media_player [ACCT-A]
value stopped
50%
handle kept
only the DARK route keeps all three
(c) where each strategy lands
leak-free
no compression
ours
published
budget cluster link
leaks
conceals
F1 ↑
0.6
0.5
0.4
0.3
0.2
0.1
0
50
100
prompt leak (%)
(d) the cost of each remedy
no compression 0.6081
DARK
0.4937
0.0% leak
compress
0.4831
98.1% leak
redact
0.1712
0.0% leak
0.0
0.2
0.4
0.6
F1 on 2,728 cases ↑